

# **Institute of Actuaries of India**

## **Subject SA4 – Pensions and Other Benefits**

### **May 2023 Examination**

#### **INDICATIVE SOLUTION**

##### **Introduction**

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

**Solution 1:**

i) Accounting valuation = AV ; regulatory valuation = RV

**Objective & purpose**

(Max 2)

- The requirement for separate valuations arises from the governing regulations. The accounting framework requires the company to prepare and report the employee benefit obligations under the accounting standard (USGAAP) in this case while the regulator requires valuation as per the principles laid down by the regulator.
- Purpose of AV is to portray a true and fair view of the obligation while the RV is prepared for regulator so as to ensure the solvency/security of the pension plan as prescribed by the regulations.
- The benefits valued, method, basis & presentation of results of the two valuations will therefore differ due to the different objectives.
- For AV, the benefits of the scheme as specified in the scheme rules/Trust deed will be valued. In case of RV, the regulatory authorities may aim to verify the security of the (minimum) level of pension benefits which will be set in the regulations.

**Method & Basis**

(Max 3)

- The AV require use of best estimate assumptions while the RV are based on prudent assumptions to ensure security of the benefits. The basis for RV are often prescribed by the regulations
  - The discount rate under the AV is based on the government bonds or corporate bonds while the discount rate under the RV is based on the yields on assets held by the trust.
  - The salary growth and pension growth under AV is best estimate assumption of expected future growth while that under RV will include an element of prudence.
  - The extent of mortality improvements if any may be different under both the valuations.
- The method of valuation for AV is PU method to reflect the accruing cost of the scheme. In case of RV, the method as prescribed by the regulations will aim to ensure the (minimum) level of funding. (For eg Aggregate, CU or any other method)

**Presentation of results & other points**

(Max 3)

- RV will typically include an action plan and recommendations for contribution rates to achieve a targeted funded status of plan as prescribed. (For eg Target funding may be 80% of discontinuance benefits)
- The AV will include a list of supporting disclosures as prescribed under the accounting standard. These include presentation of movement of the liabilities & assets, treatment of actuarial gains & losses, sensitivity of the valuation results etc. The presentation of results will often be standardized & simple to comply with. The RV may require a submission of an analysis of surplus.
- The RV will generally include recommendations on use of surplus if any. The same is not required under the AV.
- The assumptions used for RV take into consideration the sponsor covenant & quality of assets. (For eg self investment may be excluded for demonstrating funding level)
- The recommended contribution rates under RV are set to ensure buildup of fund while still not adversely impacting the sponsor's financial health.
- The frequency of valuation required under regulatory and accounting regime may be different.

[For each valid point 0.5 marks, **Max 8**]

ii) Why the existing figures are not appropriate?

(Max 2)

- The consolidated accounts post acquisition, should reflect the potential cost of the pension benefits being provided under the US company and its likely impact on the financial results of the Company. (0.5)
- The existing shareholders expect that the purchase of US company adds value to their investments. They expect that the potential risks associated with the purchase if any, is appropriately reflected in the purchase price. (1)
- The potential shareholders of the Indian company may view the purchase adversely if the expensive pension benefits are not fairly reflected in the financials. The ability of the Indian

company to raise capital in the market may be hampered if market view the purchase of the US company is not fairly reflected in the financial. (1)

- It is necessary that the consolidated accounts should fairly reflect the potential benefits of the purchase, the associated risk and the proposed measures to control the risk especially with reference to the pension benefits. (0.5)

#### How it differs from Indian Accounting Standards?

(Max 3)

- In India the listed companies are required to follow IND AS standards for reporting. Therefore the financial of the US company, its pension cost in particular should be remeasured as per Indian standards using the method & approach as prescribed under Ind AS. The presentation of the results should also conform to the standards. (1)
- The requirements under USGAAP and IND AS for employee benefits reporting are different. (0.25)
- The current set of accounting disclosures will be based on discount rates derived from corporate bonds while the discount rate under IND AS is referenced from the yields on government bonds. (1)
- The current set of disclosures may employ use of corridor approach allowed under the USGAAP. Hence there may be a portion of actuarial gains/losses that are still unrecognized. The IND AS standard does not allow this approach. (1)
- The assumptions that govern the USGAAP valuation may not hold post-acquisition. The salary growth may change if there is a change in business plan of the US entity post acquisition. (0.5)
- There may be changes in the expected attrition levels or early retirement assumptions based on the reaction of the employees to the acquisition. (0.5)
- The trustees may require change in administrative policies following the acquisition e.g. buying annuities to secure the pensions or requiring higher contributions etc. (0.5)

#### What is the alternative approach & how it works?

(Max 5)

- The Indian entity needs to re-value the scheme benefits as per provision of IND AS and based on assumptions that are expected to hold under the changing circumstances. The accounting position of the scheme will have to be evaluated as on the opening and closing date of the accounting period. (0.5)
- The Indian companies typically follow Apr to March as financial year. There may not be any assessment available for these dates and a fresh evaluation may be required.
- An assessment of expected attrition and early retirement may be based on-
  - other similar exercises by other companies in the past
  - feedback received by certain employees
  - media reactions or press releases that can indicate the general perception regarding the exercise. (1)
- The salary growth rate may be based on the revised business plan for the US entity.
- The impact of these changes on the final results of valuation will depend on the overall interaction of assumptions and also the impact on the valuation of assets. (1)
  - The discount rate is expected to reduce for the same term of obligation (the term of obligation itself may change due to change in demographic assumptions). This will lead to increase in value of obligation. The value of assets that are valued on DCF basis will increase too.
  - The early leavers may increase in case of unfavorable reaction to the acquisition. This will reduce the value of obligation and vice versa.
  - The immediate recognition of un-recognized gains/ losses will change the equity and liabilities of the US entity.
- The Company may provide information about the volatility of the cost by illustrating the sensitivity of the financial results to the changes in the parameters or in the benefit design. (1)
- There may be changes to the required contribution rates and results of regulatory valuations following the changes in circumstances. While this does not impact the results under IND AS, the Indian entity needs to be cognizant of these impacts. (0.5)
- The approach in re measuring the pension cost is to provide information to its (potential) shareholders that reflects its fair cost together with associated risk. Hence the basis & method to

measure the cost should reflect these objectives and presented as per Ind AS in consolidated accounts. (1)

[Max 10]

iii)

- The investments currently held in respect of liabilities accrued by active employees may be long term debt or equities as cash outgo are many years into the future.
- The severance plan requires that these investments be reallocated so that trust can pay transfer values or buy deferred annuities if required.
- The investment plan and contribution schedule needs to be revised completely as the fund is now closed to new entrants and moreover, the existing employees are expected to serve only for the next 5 years.
- The term of the pension liabilities becomes much shorter following closure. Hence it is necessary to match the liabilities with assets of shorter duration.
- Investment in equities & other real assets providing higher but volatile returns, needs to be re looked, as the company will be having less time to manage the deficits following closure.
- There is also a need to increase the investment in marketable, short term assets to match the short term nature of benefit purchases.
- An ALM exercise is to be conducted to assess the benefit outgo over the next 5 years & match the them with suitable assets. This may help the Trustees to avoid selling assets at time when market conditions are not favourable.
- The revised investment strategy is likely to result in lower investment returns. Hence the pension plan requires more financing due to the shift in investment policy.
- It is also necessary to monitor the movement of annuity rates in the market more closely in case the vested pensions are to be discharged immediately over the next few years.
- Trustees need to get clarity on the future management and operation of the trust with respect to the pensioners i.e. servicing the benefits through the trust vs buy-out vs settlement by way of lump-sum payments
- The entity that bears the ultimate responsibility of the pension trust after the period of 5 years needs to be ascertained. In case the US entity is responsible, the contribution plan can only extend till 5 years. On the other hand, if the parent company takes the responsibility, a contribution plan for a longer period may be made.
- Any guarantees available from the US entity or the parent company needs to be evaluated and monetary value of such guarantees need to be allowed for while estimating the future funding requirements.
- The investment policy should allow for the transfer value in respect of the 500 transferring members. Clarity needs to be obtained on the expected date of transfers and assets need to be reallocated cash or short term debt accordingly. It is not clear the extent and duration for which the trustees have placed restriction on the bulk outflows. The reduction in transfer value needs to be allowed for while making the projections.
- A lot depends on the accrued liability of the transferring employees, in case these happen to be long serving high salaried employees, thus implying a relatively high value of pension obligation, awarding a reduced transfer value may improve the funding level of the trust on buy-out basis. The investment policy can hence allow for higher number of buy-out cases.
- Investment policy should determine the allowable transfer value to the severed employees (in case they join other company with pension plan) or the extent of deferred annuities to be purchased in respect of severed employees.
- Trustees need to confirm if there is any lumpsum contribution from the parent company or the US entity as a part of severance policy. And should ensure that any such payment is used only for the severed employees and not for enhancing the funding position of the trust.
- Trustees should also confirm if the US entity/ parent company would compensate for the losses accumulated due to sale of investments at inappropriate time.
- A risk assessment needs to be performed to determine the optimum asset allocation given the objective of the trust i.e. self-administer vs buy-out
- A detailed plan needs to be prepared to achieve the target allocation taking into consideration
  - the expected contributions

- gains/losses due to sale of assets during reallocation
- taxes on above realized gains/losses or other transaction costs
- outgo in form of benefits / annuity purchase / transfer values
- need for invoking any available guarantee

[0.5 marks for each valid point, **Max 10**]

**iv) Option 1 – Lumpsum ex-gratia at the time of exit computed as transfer value + interest : (2 marks)**

Merits –

- no investment risk or mortality risk to employer
- easier to manage and administer.
- The treatment will be at par with other employees joining the company as even those employees do not get DB pension benefits.
- Portable benefit as the employees know their eligible transfer value in case they go back to US or any other company with eligible pension benefits.

De merits -

- There is no pension hence may not be welcome by the employees.
- The DB link is broken hence may not be liked by transferring members.
- The tax treatment of a non- statutory lump-sum benefit may be unfavorable for employees.

**Option 2 – Transfer IN into the DB pension benefit (with or without future accrual) : (5 marks)**

Merits –

- Employees get the defined benefit advantage that they were eligible
- Company can manage these benefits alongside the existing beneficiaries if the terms of benefits are similar or can be adapted easily.

De merits -

- The option requires re opening of the closed pension scheme. Due to this the company is re exposed to the risk of unknown cost of pension benefits for an unknown period of future time.
- Integrating the benefits of the two DB schemes on cost neutral terms will be challenging. Any changes in the design of the scheme will either likely to add cost to the company or a reduction in benefits for some categories of members.
- The transfer value received is likely to be lower as the original fund was in deficit. The Indian company will have to negotiate a lower benefit with the employees. This can become challenging especially if the incoming employees are key to business.
- The company will retain investment and mortality risk.
- The DB plan was closed 20 years ago, taking in 500 employees may alter the demographic profile of the plan (depending on the number of live members under the plan)
- Other employees may demand a DB benefit in lieu of their DC corpus.
- Lower portability - as computing the transfer value can be cumbersome in case the employees leave the company.

**Option 3 – Transfer IN into the DC pension benefit: (3 marks)**

Merits –

- Employees may benefit from higher investment returns that are typically seen in Indian markets in comparison to US markets.
- Easier for the company to manage DC scheme.
- Company does not retain investment or mortality risk.

De merits -

- Employees retain investment risk. Proper education with respect to investment choices and associated risk-rewards is essential.
- Employees unsure of the expected benefit as the amount of pension depends on the annuity rates at the time of purchase of annuity.

[For each valid point 0.5 marks, **Max 10**]

**v)**

- The present value of pension liabilities will reduce as the discount rate rises (all else remaining constant).

- The fair value of the assets will reduce too. In case the assets are listed, the rising interest rate will often depress the market value of assets and in case of assets valued using DCF, the present value will reduce with increase in discounting rate.
- The impact on the value of depends upon the type of assets held & their response to the revision in interest rates.
- The changes in the investment strategy (eg approach to profit booking) in responses to the changes in interest scenario will also impact on the value of assets.
- The extent of reduction in present values will also depend on the term of liabilities and the term of assets. Longer the term of the liabilities greater will be uncertainty.
- The impact on liabilities also depends on the value of other assumptions like pension growth and wages inflation. As the rising interest rates are often accompanied by rising inflation, the CPI is expected to grow faster hence the pension growth rate and salary growth rate assumption is likely to be revised upwards.
- The net impact on the funding status depends on far the assets & liabilities are matched & how they are likely to move in the rising interest scenario due to the global environment and the (future) management action to manage the pension plan.

#### Impact on the US pension funding status:

- The value of obligation may remain in similar range as before in case all the assumptions are revised in a way that relative impact is zero.
- The value of assets will reduce and may result in higher deficit.
- In case a higher salary growth rate is assumed, this will also increase the present value of future contributions, and hence may result in similar contribution rates as before.
- Rising interest rates may impact the prices of annuities. In case the prices of annuities reduce, the buy-out cost and hence the required contributions may go down.

#### Impact on the US pension accounting in US entity:

- The net impact of change in assumptions (discount rate, pension growth and salary growth) will be recognized immediately in statement of other comprehensive income. The same may be recognized in P&L as pension cost in case the amount of accumulated gains/losses cross the corridor limit as allowed under USGAAP.

#### Impact on the Indian entity's books:

1. The defined benefit plans as offered by Indian company (gratuity and DB pension) will be affected directly by the rising interest rates. The net impact will depend on the relative change in the underlying assumptions.
2. In case of gratuity plan, the benefits are often limited to a certain upper limit. Hence the impact of increase in salary growth may be lower than the impact due to similar increase discount rate. Hence the plan is likely to reveal net actuarial gain under rising interest rates.
3. In case of DC pension plan, the interest credited to the individual accounts will also be impacted depending upon the investment performance of the assets, the changes in the investment policy & the approach used by the Trustees in declaring interest rates under the revised economic conditions.
4. The impact on pension plan cannot be estimated as the terms of pension growth are unknown.
5. The impact of rising interest rates on Indian company's books due to the US pension plan also depend on the changes in currency. The net impact of the US pension plan determined in terms of USD may be offset or exaggerated by movement in USD-INR exchange rates.

[For each valid point 1 mark, **Max 12]**

**[50 Marks]**

#### **Solution 2:**

##### **i) EMPLOYEES' PENSION SCHEME 1995 (EPS' 95):**

(Max 1)

- Employees' Pension Scheme, 1995 (EPS) has been introduced with effect from 16.11.1995

- The membership of EPS is mandatory for employees who are members of the EPF Scheme (including Exempt Provident Funds).

Contributions: (Max 2)

- 8.33% of employees wages are contributed by the employer towards this Scheme out of the total contribution of employer to EPF of 12%. (0.5)
- There is a contribution of Government @ 1.16% in respect of each subscriber.
- The wages for contribution purpose were restricted to a maximum of INR 5,000/- from 1995 to 31st May 2001; up to a maximum of INR 6,500/- from 1st June, 2001 to 31st Aug, 2014 and up to a maximum of INR 15,000 thereafter (0.5)
- A member was allowed to make contribution to EPS without the ceiling of INR 6,500/- in which case this ceiling will not apply to the benefits also. (0.5)
- However, such option to contribute above the ceiling was applicable only until 1st Sep 2014 and only for those whose salary was above the ceiling of 6,500/- at the time of introduction of the scheme and who were contributing to the EPF scheme without the ceiling from that date itself. (0.5)

Benefits: (Max 2)

- A member shall be entitled to EPS Pension – if he has rendered the eligible service of 10 years or more and retires on attaining the age of 58 years (0.5)
- Early Pension – if he has rendered eligible service of 10 years or more and retires or otherwise ceases to be in employment before attaining the age of 58 years. (0.5)
- Computation of Pension = Pensionable salary \* Pensionable service / 70
- Where pensionable service shall mean the service for which contributions are received in respect of the employee in the Employees' Pension Fund and pensionable salary is the average of last 60 months' salary during the contribution period of service. Where the pensionable service is 20 years or more and the member superannuates on attaining the age of 58 years, a credit of 2 years is given in the Pensionable service. (0.5)
- Normally the pension is payable after attainment of 58 years irrespective of employer's policy of retirement age. However, a reduced pension after completing the age of 50 years may be paid in which case the amount of pension shall be reduced at the rate of 4% p.a. compounding for which the age falls short of 58 years. (0.5)
- The Scheme also provides pension benefits to spouse in case of death of pensioners at 50% & pro rated pension for deaths while in service. There is a provision to pay child pension also in case of such deaths (0.5)

Investment Guidelines: (Max 0.5)

- The contributions made to the Employees' Pension Fund Account is invested in accordance with the provisions of paragraph 52 of the Employees' Provident Funds Scheme, 1952.
- Net assets of the Family Pension Fund as on the 16.11.1995 merged in the Pension Fund and remain invested in the Public Account of the Government of India. The future Central Government's contribution accruing to the Pension Fund from 17th November, 1995 onwards is also invested in the Public Account of the Government of India. (0.5)

Tax Provisions: (Max 0.5)

- Contribution to EPS is tax-free, however the pension that member receives post retirement is taxable at the hands of the employee. (0.5)

[Max 6]

ii) Key impacts and pros and cons of opting for higher pension

**a) Government / EPFO / Trustees managing the EPS95 plan:**

Since EPS is a defined benefit scheme managed by the Government, any change in benefit will have to be actuarial assessed for impacts

Government will have to put up a systematic process to apply for higher pension option

Clarification should be provided to the members around the calculation of higher pension and method of calculation and deposit of money in order to apply for higher pension

Training maybe needed for various EPFO offices across the country to process applications for higher pension and be able to address members' queries

Pros:

(Max 1)

- ✓ Scheme being in news will gain more popularity
- ✓ Scheme will become more renowned as the pension amount would be relevant and will help to replace the income at retirement
- ✓ Currently, pension amount is less than INR 7500 per month which may not be relevant for most of the members
- ✓ As a social security scheme, Government would be providing better retirement protection to its members
- ✓ The scheme will be getting funds immediately as arrears of contributions of the subscribers. In case the scheme is having any liquidity problems, it will ease temporarily due to immediate inflow of funds.

Cons:

(Max 3)

- ✓ Scheme may turn out to be expensive The cost depends upon how the higher pension benefits compare with the higher contribution & the number of members exercising the option & their profile.
- ✓ The cost also will also increase in case pension benefits are paid retrospectively for pensioners.
- ✓ In case the EPS'95 fund is in deficit, the deficit will increase.
- ✓ The Government contribution to the each of the member will increase due to the revision in the definition of salary.
- ✓ In case the Scheme runs in to deficit, the Government may reluctant to increase the subsidy.
- ✓ there is a need a to update the members record, purify the errors present in the system. Any errors in the records will increase cost in terms of incorrect higher payouts or cost of redressal of grievance of a genuine subscriber who miss out the higher benefits due errors present in the system.
- ✓ Reputation risk of an incorrect decision is high as EPFO is subject to much larger public scrutiny.

[4]

**b) Employees / retired members eligible to choose this option**

With the new option in place, the eligible employees / retired members have a choice to apply for higher pension by giving up a part of their lump sum

Depending on personal circumstances, members would opt or not opt for it

Someone who may have financial liabilities to meet post retirement like repayment of loan, child marriage, etc. would prefer to not opt for higher pension

On the other hand, employees who do not want to manage a high lump sum upfront and want income to manage daily expenses from the Government may go for it

Pros:

(2 Marks)



- ✓ Under this option, there is a guaranteed lifetime pension irrespective of the movement of interest rate, annuity rate, etc. so the employee is likely to benefit
- ✓ Net replacement ratio of the employee's salary at retirement will increase as the scheme benefit will now be linked to uncapped salary, not restricted to INR 6500 or INR 15000
- ✓ EPS pension is extended to spouse as well at 50% of pension amount which will be a relevant amount on uncapped basis
- ✓ This pension amount maybe sufficient for member to meet daily expenses post retirement and there will be no risk of PF corpus running out if member lives longer as the pension is provided for the entire life
- ✓ Market volatility will not affect this pension amount

Cons:

(2 Marks)

- ✓ Government may change the pension formula or ask for higher contribution if the scheme becomes financially not viable. Member may not have a choice to claim back the PF corpus transferred.
- ✓ Retired members will have to deposit personal funds to opt for higher pension as they would have already withdrawn the PF corpus at retirement
- ✓ The scheme provides "Life pension" to the members together with spouse & child pension. If the life expectancy of member is lower than average due to health disorders, heavy smoking, etc then this option may not be suitable as pension will reduce to 50% for spouse on death of member & hence he is likely to loose out by exercising the option.
- ✓ There is no pension increases in EPS'95 benefits. Therefore there is a risk of pension being eroded in real terms due to inflation. The risk is retained by the subscriber. This risk is exacerbated for those who opt for higher pension..

[4]

c) Employers:

- ✓ There is no direct financial impact to the employers as there is no revision of the employers contributions due to the Judgement.
- ✓ In case of employers exempt from EPS'95, they will have to pay pensions on the uncapped salary. Pension cost for these employers will increase substantially.
- ✓ There will be in direct cost to them in several ways & its focus will be diverted in responding to the queries of its workforce.
- ✓ Employees /Retired employees are likely to reach out to their employer for more information, calculation, guidance on application, etc. to apply for higher pension
- ✓ Accordingly, employers should be trained to tackle the queries and help their employees. Educational sessions can be organized for the same
- ✓ Any requirement of joint declaration with employee will have to be vetted and signed off by the employer as well, adding to the operational burden on the employer.
- ✓ Any incorrect certification will likely to incur cost to the employer in terms penalty by EPFO, or disputes in the Court

[2]

iii) General

(Max 1)

The long term cost providing higher defined benefits as per the Judgement should take in to account a) the pension benefits(including family pension) payable under EPS 95 b) the arrears of contribution received due to higher contribution requirement c) the number of members likely to opt for higher pension & their profile d) expenses of implementing the Court order.

(1)

It is also necessary to understand the impact of the judgement on the overall profile of EPFO & its funds as there will be indirect cost implications on other areas as well due to the Judgement.

Cash flow approach to assess the long term cost –

Data: (Max 1)

In a Social Security Scheme, the data related to individual member maybe available but given the high volume of subscribers to EPS, it will require very high computational resources. Hence it would make sense to use model points by constructing appropriate model point with age, current service and salary.

The model points should consider all groups of subscribers who may be impacted by the judgement. These include in service members, pensioners eligible to receive higher pension benefits etc.

Assumptions: (Max 4)

In this cashflow model, critical assumptions required would be as below –

- ✓ Salary Growth rate per annum – Since the pension is now going to be uncapped, it is important to set the salary growth rate assumption to estimate the level of uncapped salary at the time of separation of each employee / model point. For pensioners there may not be any new to have salary growth assumption.
- ✓ Investment return – The contributions will be invested by Government and a certain rate of return is to be assumed in order to project the corpus at retirement. The investment returns earned by EPS'95 funds in the past may be taken as a guideline while setting investment return
- ✓ Discount rate – Discount rate may be set with reference to the Government bond of suitable term.
- ✓ Mortality assumption – should reflect the mortality experience of EPS'95 members. Since the data of EPS 95 members are huge, credible past experience of the scheme may be available & it may be used to set the mortality assumption. These assumption will require modification to allow for general improvement in longevity & the changes in the profile of EPS'95 members due to the option exercised.. Separate assumptions are to be set for in service subscribers & pensioners.
- ✓ Pre Retirement assumption – Member may apply to begin pension prior to retirement, so important to build in a decrement of early retirement assumption as well
- ✓ Disability rate – For members starting pension before normal retirement age due to disability, this assumption would be used for the valuation.
- ✓ Withdrawal rate assumption – Members who stop contributing to EPS within 10 years of service will be given back their corpus basis the factors under table “D” . Hence, this assumption is to be applied in order to estimate the number of drop-out from the Scheme.
- ✓ Age difference of member and spouse – Pension is payable to spouse post death of member at 50% of rate, however family member details may not be accurately available and may not be possible to incorporate either given the volume of subscribers. Hence, a certain assumption maybe made for spouse age difference basis the gender of the member

Methodology: (Max 4)

- ✓ A realistic valuation of the actuarial liabilities is to be carried out. The approach should be such that it does not underestimate the pension cost figures. A margin is to be kept to allow for the uncertainties.
- ✓ The model should project the existing pension benefits payable from the NRA & the enhanced pension payable following the Judgement. For family pension, the pension projections are to be done from the eligible date(eg date of death of subscriber) for each model point.
- ✓ For Active members, cost will be split in two parts – past service liability and future service liability and for retired members, cost would be computed as total service liability.
- ✓ The cash flow of pension benefits should be projected over the expected life time of members and discounted to the NRA using the set of parameters stated earlier. The difference in the pension benefits will be the likely cost of higher pension at NRA

- ✓ The contributions payable by the members during the active service under the existing & enhanced rates should be projected upto the NRA using investment assumptions & other pre retirement decrements. The difference in accumulations will give the higher pension corpus available at NRA to meet the higher pension outgo.
- ✓ The difference between the benefit outgo & the contributions at NRA will be the net cost to the EPFO at NRA for each model points.
- ✓ The net cost at NRA will be discounted to the present time using risk free discount rates to arrive at the present cost for each model point.
- ✓ The total cost to the EPFO will be computed by aggregating the cost at all model points & with the number of members available at each model points.
- ✓ An allowance is to be added to the expenses of management for implementing the court orders.

Presentation of results

(Max 2)

- ✓ It is very difficult to assess the number of members exercising the option.
- ✓ Hence the cost can be given for several proportions of options (eg 20%, 30% 50% etc)
- ✓ It is also possible to express the additional cost over the expected future service using any of the funding method.
- ✓ PU method, AA method or Aggregate method may be used depending upon the objectives of the Trustees for additional financing to fund the benefits.
- ✓ Allowance should also be given for errors in data while consolidating results. Sensitivity of the final results to the changes in parameters & to the changes in demographic profile should also be illustrated using several set of assumptions & using several different model points.

[For each valid point 0.5 marks, **Max 12]**

- iv) The model used by the financial advisor for advising individual subscriber compares the value of higher pension at the NRA against the value of contributions payable by the member. These two models, to this extent, aims to serve the objectives of the clients. But these two models differ in the following areas.

This model used by financial advisor would be very specific to an individual with actual salary details of the past, family member details, etc. as against the cash flow model described in iii) which was on aggregate level for all subscribers

The data relating to the member will be provided by the subscriber & there will no issues relating to the quality of data.

If the objective of the model is to compare the pensions & contributions at NRA, pre retirement decrements such mortality, withdrawal can be ignored.

The salary growth assumptions should reflect the individual circumstances taking into account growth prospects in the industry.

Investment return to the individual refers to the interest credit on his EPFO account. The investment return assumptions for projection should be based on historic data of the interest rates declared on EPF fund with suitable adjustments.

The post retirement pension benefits depend upon the longevity parameter used in the model. Ideally for illustrating the value of pension benefits, the financial advisor should take into account the individual health status of the member.( For eg if the individual is suffering from any terminal illness)

The allowance for expenses may be kept small in the projections.

The results of the calculation should be presented in such a way so as to enable the individual member to take informed decision on this matter.

In case the death benefits are also be compared, such comparisons should include appropriate mortality assumptions & family status specific to the individual members.

Variability of the results can be illustrated by choosing different set of parameters in the calculations.

While advising the client on this issue, the advisor should provide the following “risk warnings”. These include

- ✓ The advise is not a recommendation for opt/ or not to opt for higher pension
- ✓ Information on the risk retained should be included. (The risk of erosion in pension in real terms)
- ✓ The actual contribution requirements will vary than the figures illustrated due to the changes in interest credits & the changes in the actual salary

[For each valid point 0.5 marks, **Max 7**]

**v) Other issues apart from cost implications to be considered –**

- ✓ After assessing the cost implication, a decision is to be taken by the Trustees about implementation. In case EPFO board prefers to go for an appeal against the judgement a decision is to be taken within a limited time frame.
- ✓ If the implementation is going to cost additional expenses to the Government, the Ministry should be well informed about the additional cost.
- ✓ Board should implement the court orders within a time frame. Within this time frame, all the members of EPFO should be notified about choice available to the members following court orders. Time lines must be set for exercising the option.
- ✓ Documents such as the option form, authorisation for deducting arrears of contributions, must be framed & made available to all members.
- ✓ Must notify the additional documents to be submitted ( eg history of past contributions, consent letter given earlier etc)
- ✓ Employers must be informed about the implications of the court order and the role they are expected to play in the implementation.( certifying the past contributions of members)
- ✓ Data of EPS'95 needs to be purified on priority basis. If there any data that remain outside the system, they pose a serious financial risk to the EPFO. There will be delays in implementation if the numbers records outside the system are substantial
- ✓ Training of officials based across the country – this is a very complicated change and all PF offices should have trained staff to handle the applications and queries
- ✓ There is risk of not enough bandwidth within PF offices to address grievances / complains around the computation of pension and transfer amount of PF corpus
- ✓ Portal must be set up to upload the option forms & other documents
- ✓ Mechanism is to be established for calculating the higher benefits, additional contributions
- ✓ Mechanism is to be established for transferring the necessary contributions from the PF fund
- ✓ Separate mechanism needs to be established to process the option form of pensioners & collect the necessary lump sum arrears of contributions from them.
- ✓ Other stake holders such as income tax authorities, Trustees of EPFO should also be notified about implementation. If there are any tax & legal compliance, they must be adhered to.
- ✓ A high level committee should be set up to monitor the implementation of court orders
- ✓ Any errors or delay in implementation will have additional cost implication & additional litigation risk.

[For each valid point 0.5 marks, **Max 5**]

vi) Change in scheme design-

(Max 3)

Changes in the design include revising the accrual rate, changes in the definition of pensionable salary, limits on maximum service or maximum pension benefits, altering the vesting conditions, changes in the family/child pension etc. Increasing the retirement age etc.

The change may be implemented prospectively or retrospectively; Applicability of the changes may be set for existing members or for new members

Government –

- ✓ Change in scheme design to reduce cost would reduce the Scheme deficit and help the scheme to be financially viable.
- ✓ The impact depends upon the profile of members of EpS'95 & the changes impact different categories of members & the proposed change is implemented.
- ✓ Litigation risk may be there by aggrieved members. Clear communication of change, date from which they are effective and updated calculators maybe required to implement the same
- ✓ It is also possible for Government to change the scheme design over a period of time in a phased manner so that members approaching retirement are not significantly impacted.

Members –

- ✓ Lower benefit received than planned would affect the retirement plan of the member
- ✓ Member may regret the decision of opting for higher pension if the change affects the pension value substantially

Employer –

- ✓ There may be no direct cost implications to the employer due to the changes in the benefit design so long there is no change in the contribution. But there could be indirect implications on CTC as the retirement planning of its workforce is impacted due to the changes in design. The attrition rates, ability of the employer to recruit & retain employees will be impacted.

Increasing the contributory requirement –

(Max 3)

Government:

- ✓ This will also reduce the Scheme deficit as higher contribution will be received for providing the benefits. The impact depends upon the level of deficit & the size of the increase in contribution proposed. The impact also depends upon how the additional finance is shared between members, employers & the Government.
- ✓ Clear communication of change, date from which they are effective and who will bear the additional contribution – employee or employer need to be issued
- ✓ Tax treatment for additional contribution should also be addressed

Members –

- ✓ If members have to contribute higher, either PF contribution will go down or income in hand
- ✓ Members to pay higher than expected to receive the EPS pension

Employer –

- ✓ Employer to facilitate the transfer of additional contribution from employee to Government
- ✓ If employer is asked to contribute higher instead of employee, employer will bear higher cost & as a result CTC of employees will increase.

Increasing Equity Exposure:

(Max 2)

Government:

- ✓ Likely to earn higher return over the longer term but volatile in short term
- ✓ Should plan the investment to match term and nature of pension payout
- ✓ Scheme deficit likely to reduce over the long term if Trustees are prepared to meet the short term volatility in the investment return & th funding position.
- ✓ The impact depends upon the extent of exposure to equities and the risk mitigation mechanism in place to manage the volatility risk.

Member:

- ✓ No impact as long as there is no default of pension payout by the Government as this is a Defined Benefit Scheme

Employer:

- ✓ No impact as long as there is no change in the contributory requirements.

Improving the data quality:

(Max 2)

Government:

- ✓ Having an error free data in the system will reduce the margin/reserves for impurities.
- ✓ Asset liability matching would be easier with better data quality.
- ✓ The Trustees will be able to take more informed decision if their decisions are supported with good quality of data.
- ✓ Monitoring the progress of the funding levels will be easy through actuarial valuation if the data quality is improved.
- ✓ There is lesser chance of incorrect payments or fraud.
- ✓ Robust system around data would mean lesser queries and better administration of the scheme.
- ✓ Corrective actions will be taken in time to keep the scheme financially viable

Member:

- ✓ Better predictability of cost would help reduce the unpredictability of the scheme
- ✓ Shorter turnaround time for any queries to be answered
- ✓ Members will be given pension on time with limited documentation if robust systems are in place with correct data

Employer: Lesser number of expected queries by the Government and lesser grievances by employees.

[Each valid point 0.5 marks, **Max 10]**

**[50 Marks]**

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